

Executive Sales Dashboard

Project Overview

This project demonstrates an end-to-end Business Intelligence solution developed in Microsoft Power BI.

The objective was to transform raw transactional sales data into actionable executive-level insights through data modeling, DAX calculations, interactive visualizations, and business recommendations.

The dashboard allows decision-makers to monitor KPIs, identify top-performing customers and products, analyze regional performance, and track sales trends over time.

Business Problem

Senior management needed a dashboard capable of answering questions such as:

- Which customers generate the most revenue?
 - Which products sell the most?
 - Which categories drive overall sales?
 - Which countries contribute the highest revenue?
 - How are sales trending over time?
 - What actions should management take to improve performance?
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Tools Used

- Microsoft Power BI
 - Power Query
 - DAX
 - Data Modeling
 - Star Schema
 - Data Cleaning
 - Data Transformation
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Skills Demonstrated

- Data Cleaning
 - Data Transformation
 - Data Modeling
 - Star Schema Design
 - DAX Measures
 - KPI Development
 - Dashboard Design
 - Executive Reporting
 - Business Intelligence
 - Data Storytelling
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Dashboard KPIs

Your dashboard tracks:

- Total Sales
 - Total Customers
 - Total Quantity Sold
 - Average Order Value
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Dashboard Pages

Executive Overview

Provides an executive summary of sales performance.

Customer Analysis

Analyzes customer contribution and purchasing behavior.

Product Performance

Identifies top-selling products and categories.

Regional Analysis

Evaluates sales by country.

Business Insights

Insight 1

The United States generated over 90% of total revenue, making it the primary sales market.

Recommendation

Expand marketing campaigns in Canada and Mexico to reduce dependency on one market.

Insight 2

Technology is the highest revenue-generating category.

Recommendation

Increase inventory levels and promotional campaigns for technology products.

Insight 3

A small number of customers contribute a significant proportion of total revenue.

Recommendation

Implement VIP customer loyalty programs and personalized offers.

Insight 4

Office Supplies generated the lowest revenue.

Recommendation

Review pricing strategy and marketing effectiveness for office supplies.

Insight 5

Certain products consistently outperform others.

Recommendation

Maintain sufficient inventory for best-selling products while reviewing slow-moving inventory.

Insight 6

Sales remained relatively stable over the reporting period.

Recommendation

Introduce seasonal promotions and targeted campaigns to increase monthly growth.

Business Recommendations

Focus investment on Technology products due to their strong revenue contribution. Increase sales penetration in Canada and Mexico.

Reward high-value customers using loyalty incentives. Discontinue or redesign underperforming products.

Develop regional sales strategies rather than applying a single global strategy.

Data Modeling

Star Schema

The dashboard follows a Star Schema model consisting of one fact table and multiple dimension tables.

Fact Table

Fact Sales

Dimension Tables

- Dim Customer
- Dim Product
- Dim Calendar
- Dim Location

Benefits

- Faster query performance
 - Better scalability
 - Cleaner relationships
 - Clear DAX calculations
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Techniques Used

Include a dedicated section listing everything you applied.

- Data Cleaning using Power Query
- Removed Null Values
- Changed Data Types
- Created Calendar Table
- Built Star Schema
- One-to-Many Relationships
- Created DAX Measures
- KPI Cards
- Slicers
- Bar Charts
- Column Charts
- Line Charts
- Cross-filtering
- Dynamic Measures
- Executive Dashboard Layout
- Business Storytelling
- Insight Generation
- Decision Support Reporting